Nowell Hollier Phelps

nowellphelps.github.io | nphelps2@jh.edu | Last updated: November 2026

EDUCATION

Johns Hopkins University Baltimore, MD, USA
Doctor of Philosophy (PhD), Biostatistics 2024 – Present

- Interests: Bayesian statistics, causal inference, and machine learning and applications to global and environmental health
- Advised by Dr Abhirup Datta

University of Oxford Oxford, UK
Master of Mathematics (MMath), *Distinction* 2017 – 2021

PROFESSIONAL EXPERIENCE

African Institute for Mathematical Sciences Research and Innovation Center Kigali, Rwanda
Visiting Researcher 2024 (3.5 months)

Imperial College London London, UK
WHO Collaborating Center on NCD Surveillance, Epidemiology, and Modelling 2022 – 2024
Research Assistant in Epidemiology and Population Health

Big Data Institute, University of Oxford Oxford, UK
Summer Studentship 2021 (3 months)

PUBLICATIONS

Statistical Methodology

1. Victor PF Lhoste, Yefeng Fan, James E Bennett, Sarah Filippi, Christopher J Paciorek, Junyang Wang, Bin Zhou, Theo Rashid, **Nowell H Phelps**, Majid Ezzati. "Hierarchical Bayesian mixture model for subgroup-specific cardiometabolic phenotypes" *Journal of the Royal Statistical Society Series A* (2026)

Comparative Global Health

2. **(Co-lead author)** NCD Risk Factor Collaboration (NCD-RisC). "Obesity rise plateaus in developed nations and accelerates in developing nations", *Nature* (2026)
    - Press coverage in Reuters, Independent, Guardian, Le Monde, and more
3. NCD Risk Factor Collaboration (NCD-RisC). "Worldwide trends in diabetes prevalence and treatment from 1990 to 2022: a pooled analysis of 1108 population-representative studies with 141 million participants", *The Lancet* (2024)
4. **(Co-lead author)** NCD Risk Factor Collaboration (NCD-RisC) "Worldwide trends in the burden of underweight and obesity from 1990 to 2022: a pooled analysis of 3663 population-representative studies with 222 million school-aged children, adolescents and adults", *The Lancet* (2024)
    - >3,000 citations and press coverage in BBC, Economist, Guardian, CNN, and more
5. NCD Risk Factor Collaboration (NCD-RisC). "General and abdominal adiposity and hypertension in eight world regions: a pooled analysis of 837 population-based studies with 7.5 million participants", *The Lancet* (2024)
6. NCD Risk Factor Collaboration (NCD-RisC). "Global variation in diabetes diagnosis and prevalence based on fasting glucose and hemoglobin A1c", *Nature Medicine* (2023)
7. NCD Risk Factor Collaboration (NCD-RisC). "Diminishing benefits of urban living for children and adolescents' growth and development", *Nature* (2023)

In progress

8. **Nowell H Phelps**, Jacquie Astemborski, Kirsten Koehler, Megan Wilks, Hema C. Ramamurthi, Minkun Liu, Gregory D Kirk, Meredith C McCormack, Abhirup Datta*, Sarath Raju*. "Outdoor air pollutant exposure is associated with accelerated lung function decline in a structurally disadvantaged urban cohort" (*submitted 2026)*

* denotes equal contribution (co-lead or co-senior authors)

PRESENTATIONS

Invited Panels and Seminar

- "The obesity pandemic: trends and perspectives" *Zurich Heart House (online)* 2026
- "Understanding the global burden of obesity and the implications for cardiovascular health" *Lancet Panel on Obesity and Cardiovascular Health Outcomes, AHA Scientific Sessions, Chicago, IL, USA* 2024
- "MCMC for a large Bayesian hierarchical model" *Statistics Seminar Group, University of Kent (online)* 2023

Conference Poster

- "Outdoor air pollutant exposure is associated with accelerated lung function decline in a high-risk HIV cohort" *American Thoracic Society International Conference, Orlando, FL, USA* 2026

Departmental Presentations *Department of Biostatistics, Johns Hopkins University*

- "Creating and critiquing data visualizations for public health" *Speed Talk, Department Retreat* 2026
- "Modelling trends in population-level obesity" *BEST Seminar Series* 2026
- "Bayesian hierarchical modelling of population-level obesity" *WIT Working Group* 2025

TEACHING

Teaching Assistant *Department of Biostatistics, Johns Hopkins University*

- Statistical Methods in Public Health I-IV (MS/MPH level) 2026 – 2027
- Advanced Methods in Biostatistics III-IV (PhD level) Spring 2026
- Advanced Probability Theory I-II (PhD level) Fall 2025

Tutor *Department of Biostatistics, Johns Hopkins University*

- PhD qualifying exam on Advanced Methods in Biostatistics I-IV courses 2026

Guest Lectures *Department of Biostatistics, Johns Hopkins University*

- "Principles of data visualization", Advanced Data Science I 2026
- "Random walk models for population genetics" Advanced Probability Theory IV 2026

Workshop Instructor *Gender Equity Unit, Johns Hopkins University*

- Statistical Analysis for Gender Equity (SAGE) workshop in Kigali, Rwanda 2026 (upcoming)

SERVICE AND OUTREACH

Organizer, BLAST Working Group JHU Biostatistics 2026 – Present
Co-lead, Curriculum Committee JHU Biostatistics Student Organization 2025 – Present
Co-lead, Tea-time JHU Biostatistics Student Organization 2025 – 2026
PhD Student Mentor JHU Biostatistics Student Organization 2025 – 2026
Site Manager Opportunity Oxford Residential Program, University of Oxford 2020 & 2021

Ambassador UNIQ Summer Program, University of Oxford 2020
Team Guide International Mathematics Olympiad, Bath, UK 2019
Junior Members' Scholarship Representative Jesus College Oxford JCR 2018

HONORS AND AWARDS

Gordis Teaching Fellowship Johns Hopkins University 2026
College Prize Jesus College, University of Oxford 2021
Distinction University of Oxford, MMath Part C 2021
Summer Studentship Big Data Institute, University of Oxford 2021
Open Exhibition Jesus College, University of Oxford 2020
College Prize Jesus College, University of Oxford 2018

AFFILIATIONS

Society Memberships
American Thoracic Society
American Statistical Association

JHU Biostatistics Working Group Member
Bayesian Learning and Spatiotemporal Modeling (BLAST)
Causal Inference Working Group
Resources for Improving Statistical Education (RISE)
Wearable and Implantable Technology (WIT)